

The Transfer Matrix of the Long-Range Potts–Lattice-Gas Model in $d = 1$

1 Model and definition

We study the one-dimensional Potts–lattice-gas (PLG) with long-range power-law interactions,

$$-\beta\mathcal{H} = \sum_{r_1 \neq r_2} [J_{r_1 r_2} \delta(s_{r_1}, s_{r_2}) t_{r_1} t_{r_2} + K t_{r_1} t_{r_2} - \Delta(t_{r_1} + t_{r_2})] |r_1 - r_2|^{-a}, \quad (1)$$

where at each site i there is an occupancy variable $t_i \in \{0, 1\}$ and, when $t_i = 1$, a Potts color $s_i \in \{a, b, c\}$. The Kronecker δ equals 1 only when both sites are occupied *and* share the same color. The three couplings encode distinct physics:

- J is the ferromagnetic Potts coupling — same-color pairs are favored.
- K is a biquadratic lattice-gas coupling — occupied pairs carry an extra weight independent of color.
- Δ is the single-site chemical potential — each occupied site contributes a factor $-\Delta$.

After absorbing $|r_1 - r_2|^{-a}$ into the coupling vectors J_r, K_r, Δ_r , the pair interaction at separation $d = |r_i - r_j|$ becomes

$$-\beta h_{ij} = J_d \delta(s_i, s_j) t_i t_j + K_d t_i t_j - \Delta_d(t_i + t_j). \quad (2)$$

This is the quantity computed by the function `pair_energy(xi, xj, J[d], K[d], Delta[d])` in the code.

2 The transfer matrix $T(r)$

For any fixed distance r , the pair transfer matrix is defined as the Boltzmann weight of a single pair of sites separated by r , indexed by their joint state:

$$T_r(x_i, x_j) = \exp[-\beta h_{ij}] = \exp[J_r \delta(s_i, s_j) t_i t_j + K_r t_i t_j - \Delta_r(t_i + t_j)]. \quad (3)$$

Each index $x \in \{0, a, b, c\}$ runs over the 4 single-site states: $x = 0$ for the vacant site, $x \in \{a, b, c\}$ for the three Potts colors of an occupied site. Thus $T(r)$ is 4×4 .

Enumerating the four cases gives an explicit form with only *four* distinct entries, listed below.

Case	$t_i t_j$	$\delta(s_i, s_j)$	$\log T_r(x_i, x_j)$
both vacant	0	—	0
one occupied, one vacant	0	—	$-\Delta_r$
both occupied, same color	1	1	$J_r + K_r - 2\Delta_r$
both occupied, different colors	1	0	$K_r - 2\Delta_r$

3 Analytic form

In the ordering $\{0, a, b, c\}$ for rows and columns, the transfer matrix is

$$T(r) = \begin{pmatrix} 1 & e^{-\Delta_r} & e^{-\Delta_r} & e^{-\Delta_r} \\ e^{-\Delta_r} & e^{J_r+K_r-2\Delta_r} & e^{K_r-2\Delta_r} & e^{K_r-2\Delta_r} \\ e^{-\Delta_r} & e^{K_r-2\Delta_r} & e^{J_r+K_r-2\Delta_r} & e^{K_r-2\Delta_r} \\ e^{-\Delta_r} & e^{K_r-2\Delta_r} & e^{K_r-2\Delta_r} & e^{J_r+K_r-2\Delta_r} \end{pmatrix}. \quad (4)$$

The matrix is symmetric (detailed balance) and invariant under any permutation of the color labels $\{a, b, c\}$ (Potts S_3 symmetry).

Normalization used in the code. `construct_transfer_matrix(J, K, Delta, r, normalize=True)` returns $T(r)$ divided by its largest entry. Whenever $J_r + K_r > 0$ and Δ_r is not too large and negative, the largest entry is the same-color diagonal block, $e^{J_r+K_r-2\Delta_r}$, and the normalized matrix reads

$$T^{\text{norm}}(r) = \begin{pmatrix} e^{-(J_r+K_r-2\Delta_r)} & e^{-(J_r+K_r-\Delta_r)} & e^{-(J_r+K_r-\Delta_r)} & e^{-(J_r+K_r-\Delta_r)} \\ e^{-(J_r+K_r-\Delta_r)} & 1 & e^{-J_r} & e^{-J_r} \\ e^{-(J_r+K_r-\Delta_r)} & e^{-J_r} & 1 & e^{-J_r} \\ e^{-(J_r+K_r-\Delta_r)} & e^{-J_r} & e^{-J_r} & 1 \end{pmatrix}. \quad (5)$$

In this form, the ratio between the on- and off-diagonal color entries is e^{J_r} , a clean readout of the Potts coupling alone.

4 Graphical layout of the four blocks

The 4×4 matrix has four symmetry-distinct entry types, colored here by block:

	0	a	b	c
0	1	$e^{-\Delta_r}$	$e^{-\Delta_r}$	$e^{-\Delta_r}$
a	$e^{-\Delta_r}$	$e^{J_r+K_r-2\Delta_r}$	$e^{K_r-2\Delta_r}$	$e^{K_r-2\Delta_r}$
b	$e^{-\Delta_r}$	$e^{K_r-2\Delta_r}$	$e^{J_r+K_r-2\Delta_r}$	$e^{K_r-2\Delta_r}$
c	$e^{-\Delta_r}$	$e^{K_r-2\Delta_r}$	$e^{K_r-2\Delta_r}$	$e^{J_r+K_r-2\Delta_r}$

Legend: vacant–vacant vacant–occupied same color different colors

5 Physical interpretation of each coupling

Each of J_r , K_r , Δ_r controls a distinct family of entries, which makes the matrix easy to read.

J_r — Potts ferromagnetic coupling

J_r appears *only* in the same-color diagonal of the occupied–occupied block (amber cells). It lifts same-color entries above different-color entries by exactly a factor e^{J_r} . A large positive J_r means

the color symmetry is spontaneously broken: along the one-dimensional chain, long stretches of a single color are favored.

K_r — vacancy-vacancy biquadratic

K_r appears in every occupied–occupied entry (amber and purple). It shifts the relative weight of “both sites occupied” vs. other cases. A positive K_r favors density; a negative K_r favors vacancy. In the RG flow, K_r typically grows from zero because decimation generates effective attractions between occupied sites even starting from bare $K_0 = 0$. (Hence three coupling families are needed for a closed RG recursion; dropping K would not close.)

Δ_r — chemical potential

Each occupied site contributes a factor $e^{-\Delta_r}$ additively in the exponent. Occupied–occupied entries carry $-2\Delta_r$, occupied–vacant entries carry $-\Delta_r$, and vacant–vacant carries none. Large positive Δ_r suppresses occupancy and drives the system to the dilute phase; large negative Δ_r drives it to full occupancy.

6 Signatures of the phase sinks

Reading the *normalized* transfer matrix $T^{\text{norm}}(r)$ at the end of a long RG flow is a direct diagnostic of which fixed point has been reached. The heuristic classifier `determine_phase_from_TM` in `renormalization_plg.py` is based on this table:

Phase sink	Normalized-matrix signature	Coupling limits
Potts-ordered, dense	diag. color block ≈ 1 , off-diag. color ≈ 0 , vac. row ≈ 0	$J_r \rightarrow \infty, \Delta_r \rightarrow -\infty$
Disordered, dense	all occupied–occupied entries ≈ 1 , vac. row ≈ 0	$J_r \rightarrow 0, K_r \text{ finite}, \Delta_r \rightarrow -\infty$
Disordered, dilute	$T_{00} \approx 1$, all other entries ≈ 0	$J_r, K_r \rightarrow 0, \Delta_r \rightarrow +\infty$

Remark on the trivial fixed point. Under the majority-rule cell projection used in the RG, the point $J = K = \Delta = 0$ is *not* a fixed point, because the projection favors “cell occupied” over “cell vacant” purely by entropy (there are many more microscopic realizations of an occupied cell than of a vacant one). Applying one RG step to the all-zero coupling vector yields $\Delta' = \log(R(00)/R(a0)) < 0$; this is the decimation’s drift toward a dense, disordered state even before any physical interaction is introduced. The true high-temperature disordered fixed point of this RG therefore lies at some finite negative Δ^* , not at the origin.

7 Relation to the decimation output

The same four entry types of $T(r)$ reappear, enlarged, in the two-cell decimation over a 6-spin cluster: the four Boltzmann sums $R_{r'}(00)$, $R_{r'}(a0)$, $R_{r'}(ab)$, $R_{r'}(aa)$ are exactly the cell-level analogs of the four colored blocks in Eq. (4), and the renormalized couplings are extracted via

$$J'_{r'} = \log R_{r'}(aa) - \log R_{r'}(ab), \quad (6)$$

$$K'_{r'} = \log R_{r'}(ab) + \log R_{r'}(00) - 2 \log R_{r'}(a0), \quad (7)$$

$$\Delta'_{r'} = \log R_{r'}(00) - \log R_{r'}(a0). \quad (8)$$

These relations are structurally identical to reading the four distinct values of $T(r)$ and inverting Eq. (4) for (J_r, K_r, Δ_r) — the only difference is that the decimation sums over internal configurations to produce the *cell-level* Boltzmann weights.